Project Proposal

Automated Triage of Protein Crystallisation Images
for a Synchrotron-Facility-Style Crystallography Workflow

Student name	Chamika Subashinie
Student ID	COMScDS25.2P-001
Programme / module	MSc Data Science – AI Coursework (Specialised Artificial Neural Networks)
Lecturer	Dr. Sumani Rajadurai
Date	4 October 2026
Submission deadline	6 November 2026

1. Problem Statement

In macromolecular crystallography, protein samples are set up in crystallisation plates and imaged repeatedly over days or weeks. Only drops that contain crystals are worth harvesting and sending to a synchrotron beamline for X-ray diffraction. Successful crystallisation is rare, and crystallographers still inspect very large numbers of drop images by eye. This project develops an artificial neural network that classifies drop images and flags those likely to contain crystals.

The system is designed to fit a synchrotron-facility-style macromolecular crystallography workflow, at the crystallisation-screening stage that precedes beamline data collection. It follows publicly documented open-source software conventions (e.g. the DiamondLightSource python-copier-template).

2. Intended User and Decision Supported

The user of this system is	a crystallographer or crystallisation-facility scientist reviewing imaged crystallisation plates.
When the model produces a prediction, the user	prioritises drops flagged as likely to contain crystals for close inspection and harvesting for beamline data collection, and deprioritises the rest for manual review. The system supports the decision; it does not discard drops automatically.

3. Decision Consequences and Primary Metric

False negative (a missed crystal): a hit produced from scarce or expensive protein may be lost. **False positive**: a few minutes of unnecessary inspection. The errors are asymmetric and the Crystals class is a minority class. The proposed primary metric is **recall on the Crystals class at a fixed minimum precision**, with **PR-AUC for Crystals** as a supporting metric. Overall accuracy is unsuitable because it is dominated by the majority classes. The final metric definition will be justified in the EDA, and the decision threshold will be chosen on validation data only.

4. Dataset

Name	MARCO – MAchine Recognition of Crystallization Outcomes
Source	University at Buffalo, Center for Computational Research – https://marco.ccr.buffalo.edu
Licence	Creative Commons Attribution 4.0 International (CC BY 4.0)
Size	462,804 expert-scored images (415,775 published training; 47,029 published test)
Contributing sources	CSIRO Collaborative Crystallisation Centre, GSK, Hauptman-Woodward Medical Research Institute, Merck, Bristol-Myers Squibb; imaged on Rigaku, Formulatrix and in-house systems
Target	Four expert labels: Clear, Crystals, Precipitate, Other. Crystals (about 13% of the training portion) is the decision-relevant class.
Modality	Images (unstructured data)

Format	TFRecords, larger than 500 MB; a download-and-extraction script and a representative sample will be submitted
Personal data	None – images show crystallisation experiments only
Suitability	Not on the excluded list. Prior research exists (Bruno et al., 2018) and will be cited, but no end-to-end tutorial answers this coursework.

5. Data Preparation, Split and Leakage Control

- Stratified subsample of approximately 30,000–40,000 images, stratified by label and source institution, to keep training feasible.
- One fixed train/validation/test split, used for the classical baseline, all ANN variants and the GenAI evaluation. MARCO's published split will not be reused, because it may contain near-duplicates across its boundary.
- The metadata has no drop or plate ID, and drops may be imaged several times. Filenames will be checked for plate/well information, and perceptual-hash near-duplicate detection will group related images so they stay on one side of the split.
- The test set will be used once only, for final evaluation. Random seeds will be fixed and reported.

6. Proposed Approach by Assessed Question

Question	Planned work
Q1 Dataset and EDA	Class and source distributions, image size and quality checks, near-duplicate analysis, leakage checks, metric justification, three modelling-constraining findings.
Q2 Solution design	Fine-tuned pretrained CNN backbone with a class-weighted loss reflecting the error asymmetry. Alternatives discussed: training from scratch, vision transformers, frozen-feature classifiers, classical feature pipelines.
Q3 Development and optimisation	Classical baseline (HOG/LBP/texture features with tuned gradient boosting or SVM) at comparable tuning effort. Trivial benchmark, initial and final ANN, learning curves, error analysis by class and source. Study of at least three hyperparameters and two optimisers (e.g. SGD with momentum, AdamW). Full experiment log including failed runs.
Q4 Web application	Frontend and backend; the user uploads a drop image and the ANN predicts at request time. Class scores shown with a plain explanation of what they represent, stated limitations, and tests with normal, invalid and out-of-distribution inputs.
Q5 Explainable AI	Grad-CAM in the application, compared with occlusion sensitivity and a model-agnostic method (LIME or SHAP). Sanity checks by occluding highlighted regions; analysis of incorrect predictions.
Q6 Generative AI	Zero-shot classification of the same test images with a vision-language model, same primary metric, prompts selected on validation data only. Local open-weights and hosted API models compared on performance, latency, cost and privacy.

References

Bruno, A. E. et al. (2018) ‘Classification of crystallization outcomes using deep convolutional neural networks’, *PLOS ONE*, 13(6), e0198883.

MARCO dataset. University at Buffalo. Available at: https://marco.ccr.buffalo.edu (CC BY 4.0).

Diamond Light Source. python-copier-template. Available at: https://github.com/DiamondLightSource/python-copier-template.